

BANK ACCOUNT MANAGEMENT SYSTEM IN C

Mini Project Report

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Objective

The objective of this project is to develop a simple Bank Management System using the C programming language. The project performs basic banking operations such as:

- Create Account
- Deposit Money
- Withdraw Money
- Balance Enquiry
- Exit

This project demonstrates the use of:

- Structures
- Functions
- Arrays
- Menu Driven Programming

Introduction

A Bank Management System is used to store customer account details and perform banking operations. This mini-project helps beginners understand how real banking applications work by utilizing basic C programming concepts.

Features

1. Create New Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit

Algorithm

1. Start the program.
2. Display the menu options.
3. Read the user choice.
4. Perform the selected banking operation.
5. Repeat until the user selects Exit.
6. Stop the program.

Program Code

```
#include <stdio.h>
#include <stdlib.h>

struct Account
{
    int accNo;
    char name[50];
    float balance;
};
```

```
struct Account acc[100];
```

```
int count = 0;
```

```
// Function Declarations
```

```
void createAccount();
```

```
void deposit();
```

```
void withdrawMoney();
```

```
void balanceEnquiry();
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    while (1)
```

```
    {
```

```
        printf("\n=====\\n");
```

```
        printf("    BANK MANAGEMENT SYSTEM\\n");
```

```
        printf("=====\\n");
```

```
        printf("1. Create Account\\n");
```

```
        printf("2. Deposit Money\\n");
```

```
        printf("3. Withdraw Money\\n");
```

```
        printf("4. Balance Enquiry\\n");
```

```
        printf("5. Exit\\n");
```

```
        printf("Enter your choice: ");
```

```
        scanf("%d", &choice);
```

```
        switch (choice)
```

```
{  
    case 1:  
        createAccount();  
        break;  
  
    case 2:  
        deposit();  
        break;  
  
    case 3:  
        withdrawMoney();  
        break;  
  
    case 4:  
        balanceEnquiry();  
        break;  
  
    case 5:  
        printf("Thank You!\n");  
        exit(0);  
  
    default:  
        printf("Invalid Choice!\n");  
}  
}  
  
return 0;  
}
```

```
void createAccount()
{
    printf("\nEnter Account Number: ");
    scanf("%d", &acc[count].accNo);

    printf("Enter Name: ");
    scanf(" %[^\\n]", acc[count].name);

    printf("Enter Initial Balance: ");
    scanf("%f", &acc[count].balance);

    count++;

    printf("Account Created Successfully!\\n");
}
```

```
void deposit()
{
    int accNo, i, found = 0;
    float amount;

    printf("\nEnter Account Number: ");
    scanf("%d", &accNo);

    printf("Enter Amount to Deposit: ");
    scanf("%f", &amount);

    for (i = 0; i < count; i++)
    {
```

```
    if (acc[i].accNo == accNo)
    {
        acc[i].balance += amount;

        printf("Deposit Successful!\n");
        printf("Updated Balance: %.2f\n", acc[i].balance);

        found = 1;
        break;
    }
}

if (!found)
{
    printf("Account Not Found!\n");
}
}

void withdrawMoney()
{
    int accNo, i, found = 0;
    float amount;

    printf("\nEnter Account Number: ");
    scanf("%d", &accNo);

    printf("Enter Amount to Withdraw: ");
    scanf("%f", &amount);
```

```
for (i = 0; i < count; i++)
{
    if (acc[i].accNo == accNo)
    {
        if (amount > acc[i].balance)
        {
            printf("Insufficient Balance!\n");
        }
        else
        {
            acc[i].balance -= amount;

            printf("Withdrawal Successful!\n");
            printf("Remaining Balance: %.2f\n", acc[i].balance);
        }

        found = 1;
        break;
    }
}

if (!found)
{
    printf("Account Not Found!\n");
}
}
```

```
void balanceEnquiry()
```

```
{
```



```
int accNo, i, found = 0;

printf("\nEnter Account Number: ");
scanf("%d", &accNo);

for (i = 0; i < count; i++)
{
    if (acc[i].accNo == accNo)
    {
        printf("\n===== ACCOUNT DETAILS =====\n");
        printf("Account Number : %d\n", acc[i].accNo);
        printf("Name          : %s\n", acc[i].name);
        printf("Balance       : %.2f\n", acc[i].balance);

        found = 1;
        break;
    }
}

if (!found)
{
    printf("Account Not Found!\n");
}
}
```

Output

The screenshot shows the Programiz Online C Compiler interface. The code in `main.c` defines a `BANK MANAGEMENT SYSTEM` menu with options: 1. Create Account, 2. Deposit Money, 3. Withdraw Money, 4. Balance Enquiry, and 5. Exit. The program prompts the user to enter their choice. When choice 1 is selected, it asks for an account number (45), name (Rohit Sharma), and initial balance (454545). It then prints "Account Created Successfully!". The output window on the right shows the same sequence of prompts and the successful creation message.

```
main.c
159 {
160     int accNo, i, found = 0;
161
162     printf("\nEnter Account Number: ");
163     scanf("%d", &accNo);
164
165     for (i = 0; i < count; i++)
166     {
167         if (acc[i].accNo == accNo)
168         {
169             printf("\n==== ACCOUNT DETAILS =====\n");
170             printf("Account Number : %d\n", acc[i].accNo);
171             printf("Name           : %s\n", acc[i].name);
172             printf("Balance          : %.2f\n", acc[i].balance);
173
174             found = 1;
175             break;
176         }
177     }
178
179     if (!found)
180     {
181         printf("Account Not Found!\n");
182     }
183 }
```

Output

```
=====
BANK MANAGEMENT SYSTEM
=====
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit
Enter your choice: 1

Enter Account Number: 45
Enter Name: Rohit Sharma
Enter Initial Balance: 454545
Account Created Successfully!

=====
BANK MANAGEMENT SYSTEM
=====
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit
Enter your choice:
```

Account Created Successfully

The screenshot shows the Programiz Online C Compiler interface. The code in `main.c` is the same as the previous one, but the output window shows the next step in the program. After the account creation, the user is prompted to enter their choice again. When choice 2 is selected, it asks for an account number (45) and an amount to deposit (3333). It then prints "Deposit Successful!" and "Updated Balance: 457878.00". The output window on the right shows the same sequence of prompts and the successful deposit message.

```
main.c
159 {
160     int accNo, i, found = 0;
161
162     printf("\nEnter Account Number: ");
163     scanf("%d", &accNo);
164
165     for (i = 0; i < count; i++)
166     {
167         if (acc[i].accNo == accNo)
168         {
169             printf("\n==== ACCOUNT DETAILS =====\n");
170             printf("Account Number : %d\n", acc[i].accNo);
171             printf("Name           : %s\n", acc[i].name);
172             printf("Balance          : %.2f\n", acc[i].balance);
173
174             found = 1;
175             break;
176         }
177     }
178
179     if (!found)
180     {
181         printf("Account Not Found!\n");
182     }
183 }
```

Output

```
Enter your choice: 1

Enter Account Number: 45
Enter Name: Rohit Sharma
Enter Initial Balance: 454545
Account Created Successfully!

=====
BANK MANAGEMENT SYSTEM
=====
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit
Enter your choice: 2

Enter Account Number: 45
Enter Amount to Deposit: 3333
Deposit Successful!
Updated Balance: 457878.00

=====
BANK MANAGEMENT SYSTEM
=====
```

Deposit Successful

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```
main.c
159 {
160     int accNo, i, found = 0;
161
162     printf("\nEnter Account Number: ");
163     scanf("%d", &accNo);
164
165     for (i = 0; i < count; i++)
166     {
167         if (acc[i].accNo == accNo)
168         {
169             printf("\n==== ACCOUNT DETAILS =====\n");
170             printf("Account Number : %d\n", acc[i].accNo);
171             printf("Name : %s\n", acc[i].name);
172             printf("Balance : %.2f\n", acc[i].balance);
173
174             found = 1;
175             break;
176         }
177     }
178
179     if (!found)
180     {
181         printf("Account Not Found!\n");
182     }
183 }
```

Output

```
=====
BANK MANAGEMENT SYSTEM
=====
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit
Enter your choice: 3

Enter Account Number: 45
Enter Amount to Withdraw: 3333
Withdrawal Successful!
Remaining Balance: 454545.00

=====
BANK MANAGEMENT SYSTEM
=====
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit
Enter your choice: |
```

Withdrawal Successful

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```
main.c
159 {
160     int accNo, i, found = 0;
161
162     printf("\nEnter Account Number: ");
163     scanf("%d", &accNo);
164
165     for (i = 0; i < count; i++)
166     {
167         if (acc[i].accNo == accNo)
168         {
169             printf("\n==== ACCOUNT DETAILS =====\n");
170             printf("Account Number : %d\n", acc[i].accNo);
171             printf("Name : %s\n", acc[i].name);
172             printf("Balance : %.2f\n", acc[i].balance);
173
174             found = 1;
175             break;
176         }
177     }
178
179     if (!found)
180     {
181         printf("Account Not Found!\n");
182     }
183 }
```

Output

```
=====
BANK MANAGEMENT SYSTEM
=====
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit
Enter your choice: 4

Enter Account Number: 45

==== ACCOUNT DETAILS ====
Account Number : 45
Name : Rohit Sharma
Balance : 454545.00

=====
BANK MANAGEMENT SYSTEM
=====
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit
```

Balance Displayed

The screenshot shows a web browser with the URL `programiz.com/c-programming/online-compiler/`. The page title is "Programiz C Online Compiler". The code editor on the left contains the following C code:

```
main.c
159 {
160     int accNo, i, found = 0;
161
162     printf("\nEnter Account Number: ");
163     scanf("%d", &accNo);
164
165     for (i = 0; i < count; i++)
166     {
167         if (acc[i].accNo == accNo)
168         {
169             printf("\n==== ACCOUNT DETAILS =====\n");
170             printf("Account Number : %d\n", acc[i].accNo);
171             printf("Name : %s\n", acc[i].name);
172             printf("Balance : %.2f\n", acc[i].balance);
173
174             found = 1;
175             break;
176         }
177     }
178
179     if (!found)
180     {
181         printf("Account Not Found!\n");
182     }
183 }
```

The output window on the right shows the following text:

```
3. Withdraw Money
4. Balance Enquiry
5. Exit
Enter your choice: 4

Enter Account Number: 45

==== ACCOUNT DETAILS ====
Account Number : 45
Name : Rohit Sharma
Balance : 454545.00

=====
BANK MANAGEMENT SYSTEM
=====
1. Create Account
2. Deposit Money
3. Withdraw Money
4. Balance Enquiry
5. Exit
Enter your choice: 5
Thank You!

=== Code Execution Successful ===
```

Conclusion

The Bank Account Management System project was successfully developed using the C programming language. This project demonstrates the implementation of basic banking operations such as account creation, deposit, withdrawal, and balance enquiry using structures and functions. It also helps in understanding menu-driven programming, arrays, loops, and conditional statements. The system is simple, user-friendly, and suitable for beginners to learn fundamental concepts of C programming and real-time application development.